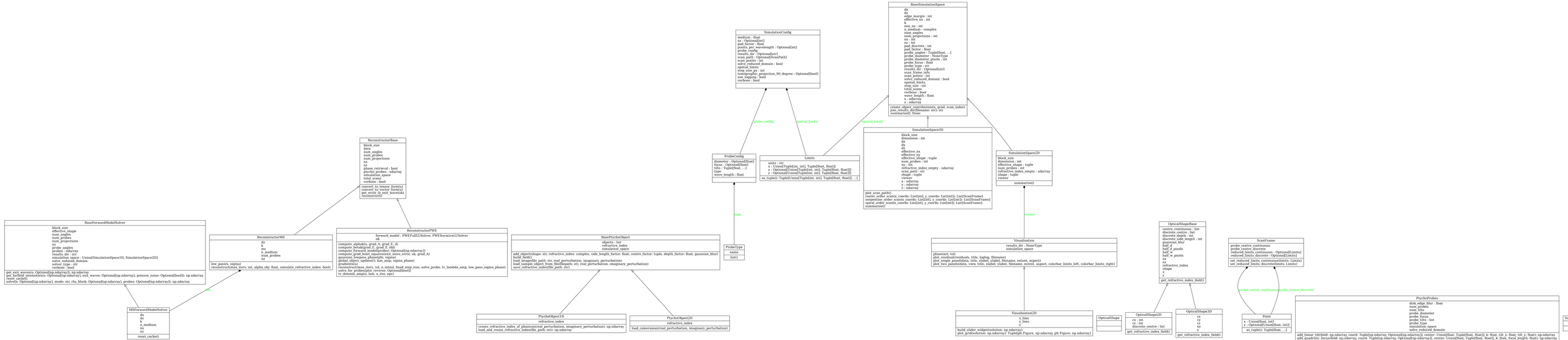
BaseSimulationSpace
dx
dz
edge_margin : int
effective_nx : int
n
min_nx : int
n_medium : complex
num_angles
num_projections : int
nx : int
nz : int
pad discrete : int
pad_factor : float
probe_angles : Tuple[float, ...]
probe diameter : NoneType
probe_diameter_pixels : int
probe_focus : float
probe_tilts : list
results_dir : Optional[str]
scan_frame_info
scan_points : int
solve_reduced_domain : bool
spatial_limits
step_size : int
total_scans
use_logging : bool
wave length : float
x : ndarray
z : ndarray
create_object_contribution(n, grad, scan_index)
join_results(filename: str) str
summarise() None
SimulationConfig
medium : float
num_iterations : int
pad factor : float
points_per_wavelength : Optional[int]
probe_config
results_dir : Optional[str]
scan_frame : Optional[ScanPath]
scan_points : int
solve_reduced_domain : bool
spatial_limits
step_size px : int
tomographic_projection_90_degree : Optional[bool]
use_logging : bool
verbose : bool
probe_config
spatial_limits
spatial_limits
ProbeConfig
diameter : Optional[float]
focus : Optional[float]
tilts : Tuple[float, ...]
type
wave_length : float
Limits
units : str
x : Union[Tuple[int, int], Tuple[float, float]]
y : Optional[Union[Tuple[int, int], Tuple[float, float]]]
z : Optional[Union[Tuple[int, int], Tuple[float, float]]]
as_tuple() : Tuple[Union[Tuple[int, int], Tuple[float, float]], ...]
SimulationSpace3D
block_size
dimension : int
dx
dy
dz
effective_nx
effective_ny
effective_shape : tuple
num_probes : int
ny : int
refractive_index_empty : ndarray
scan_path : str
shape : tuple
viewer
x : ndarray
y : ndarray
z : ndarray
plot_scan_path()
plot_spiral_order_scan(x_coords: List[int], y_coords: List[int]): List[ScanFrame]
serpentine_order_scan(x_coords: List[int], y_coords: List[int]): List[ScanFrame]
spiral_order_scan(x_coords: List[int], y_coords: List[int]): List[ScanFrame]
summarise()
SimulationSpace2D
block_size
dimension : int
effective_shape : tuple
num_probes : int
refractive_index_empty : ndarray
shape : tuple
viewer
summarise()
type
ProbeType
name
list()
viewer
ReconstructorBase
block_size
dimension
dx
num_angles
num_probes
num_projections
nx
nz
phase_retrieval : bool
ptycho_probes : ndarray
simulation_space
solver_type
verbose : bool
convert_to_tensor_form(x)
convert_to_vector_form(x)
get_error_in_exit_wave(x)
reconstruct()
BaseForwardModelSolver
block_size
dimension : int
num_angles
num_probes
num_projections
nx
probe_angles
probes : ndarray
results_dir : str
simulation_space : Union[SimulationSpace1D, SimulationSpace2D]
solve_reduced_domain
solver_type : str
verbose : bool
get_exit_waves(x: Optional[np.ndarray]): np.ndarray
get_farfield_intensities(x: Optional[np.ndarray]), exit_waves: Optional[np.ndarray], poisson_noise: Optional[bool]): np.ndarray
view_coords()
waves(x: Optional[np.ndarray], mode: str, rhs_block: Optional[np.ndarray], probes: Optional[np.ndarray]): np.ndarray
ReconstructorMS
dz
k
nx
n_medium
num_probes
nz
low_pass(x, sigma)
reconstruct(max_iters: int, alpha_obj: float, simulate_refractive_index: bool)
ReconstructorPWE
forward_model : PWEFullLUSolver, PWEIterativeLUSolver
compute_alpha(x, grad_A, grad_E, s)
compute_beta(grad_E, grad_E_old)
compute_forward_model(probes: Optional[np.ndarray])
compute_grad_least_squares(exit_wave error, uk, grad_A)
compute_grad_regularisation(uk)
global_object_update(t, lam_amp, sigma_phase)
gradients(u)
reconstruct(max_iters, tol, n_initial, fixed_step_size, solve_probe, tv_lambda_amp, low_pass_sigma_phase)
solve_for_probes(plot_reverse: Optional[bool])
tv_denoise_amp(u, lam, n_iter, eps)
BasePtychoObject
objects : list
refractive_index
simulation_space
add_objects(shape: str, refractive_index: complex, side_length_factor: float, centre_factor: tuple, depth_factor: float, gaussian_blur: bool)
load_image(file_path: str, real_perturbation, imaginary_perturbation)
load_sample_object_from_files(filepath: str, real_perturbation, imaginary_perturbation)
save_refractive_index(file_path: str)
Visualisation
results_dir : NoneType
simulation_space
phasestarry, tol)
plot_pseudocolouridsuals, title, logfog, filename)
plot_single_panel(data, title, xlabel, ylabel, filename, extent, aspect)
plot_two_panels(data, view, title, xlabel, ylabel, filename, extent, aspect, colorbar_limits left, colorbar_limits right)
OpticalShapeBase
centre continuous : list
discrete_centre : list
discrete depth : int
discrete side_length : int
gaussian_blur
half_d
half_d_pixels
half_w
half_w_pixels
nx
nz
refractive_index
shape
x
z
get_refractive_index_field()
ScanFrame
probe_centre continuous
probe_centre_discrete
reduced limits_continuous : Optional[Limits]
reduced_limits_discrete : Optional[Limits]
set_reduced_limits_continuous(limits: Limits)
set_reduced_limits_discrete(limits: Limits)
probe_centre_continuous
probe_centre_discrete
MSForwardModelSolver
dx
dz
k
n_medium
nx
nz
reset_cache()
ref
PtychoObject1D
refractive_index
create_refractive_index_of_phantom(real_perturbation, imaginary_perturbation): np.ndarray
load_and_resize_refractive_index(file_path: str): np.ndarray
PtychoObject2D
refractive_index
load_cameraman(real_perturbation, imaginary_perturbation)
Visualisation2D
x_bins
y_bins
z
build_slider_widget(solution: np.ndarray)
plot_grid(solution: np.ndarray): Tuple[plt.Figure, np.ndarray, plt.Figure, np.ndarray]
OpticalShape
OpticalShape2D
cx : int
cz : int
discrete_centre : list
get_refractive_index_field()
OpticalShape3D
cx
cy
cz
y
get_refractive_index_field()
Point
x : Union[float, int]
y : Optional[Union[float, int]]
as_tuple() : Tuple[float, ...]
PtychoProbes
disk_edge_blur : float
num_probes
probe_focus
probe_diameter
probe_focus
probe_tilts : list
probe_type
simulation_space
solve_reduced_domain
add_linear_tilt(field: np.ndarray, coord: Tuple[np.ndarray, Optional[np.ndarray]], center: Union[float, Tuple[float, float]], k: float, tilt_x: float, tilt_y: float): np.ndarray
add_quadratic_focus(field: np.ndarray, coord: Tuple[np.ndarray, Optional[np.ndarray]], center: Union[float, Tuple[float, float]], k: float, focal_length: float): np.ndarray
build_probes() np.ndarray
create_empty_probe_stack()
make_single_probe(scan: int, probe_type: Optional[str], probe_focus: Optional[float], probe_tilt: Optional[float]): np.ndarray
ScanPath
name
list()